

Contribution Statement

ChartPRM: Process Reward Modeling and Step-Level Preference Optimization for Complex Multimodal Chart Reasoning

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Both team members contributed meaningfully to this project. Responsibilities and completed tasks are listed below.

Yahor Lahunovich

Built the core ChartPRM pipeline on **CharXiv** and authored the majority of the report.

- Curated and stratified the 500-question CharXiv reasoning pool (8 domains, 62 chart types) and the 100-question holdout.
- Built the structured chain-of-thought rollout generation protocol for Qwen2.5-VL-3B-Instruct.
- Integrated the muse-spark-1.1 multimodal LLM-as-judge with rollout-level batching.
- Built the preference-dataset curation and training pipelines for SFT, Full DPO, Step-DPO, KTO, and SFT→DPO.
- Ran and evaluated all five systems on the CharXiv holdout.
- Built the visualization and figure-generation system used throughout the report.
- Co-authored and revised the project report throughout, including writing, structure, and the bibliography.

[Yahor – please revise this to your own words and add your own AI-assistance disclosure, if any (.cursorrules/.agents/AGENTS.md list Cursor, Gemini, Claude, and AntigraVity as intended tools from early on).]

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Presented Yin et al.'s *Dynamic and Generalizable Process Reward Modeling* (DG-PRM) for the seminar and directed its adaptation, plus a cross-dataset study.

- Directed the reward-tree construction: 33 scored criteria under 9 categories, from clustered judge critiques.
- Directed the dynamic multi-criteria judge (scored via gemini-3.5-flash-lite, distinct from the muse-spark-1.1 binary judge) and Pareto-dominance-filtered pair selection for DPO (**Pareto-DPO**).
- Directed the reference-free **SimPO** baseline implementation.
- Directed the ChartGemma specialist-baseline and oracle-ceiling diagnostic experiments.
- Designed and ran the cross-dataset generalization test on **ChartQA**: sampling, rollout generation, unmodified reward-tree transfer, and retraining/evaluating all seven methods on a 30-question holdout.
- Authored the DynamicPRM, Test-Time Verification and Search, and Cross-Dataset Generalization (ChartQA) sections, and their tables/figures; also revised the Abstract, Introduction, and ChartPRM Framework as the project's scope grew to two benchmarks and seven methods.
- **AI usage summary:** Used **Claude Code** (Anthropic) as an AI coding agent for essentially all of the code and analysis above (reward tree, dynamic judge, SimPO, ChartQA pipeline) and for drafting my report sections. I directed every experiment (what to try, when to pivot), set the evidentiary bar (e.g., requiring bootstrap significance tests before claiming a win), and personally verified and interpreted all results before they went into the report.

Both authors confirm this statement accurately reflects their respective contributions to the ChartPRM project.